

Network Troubleshooting Using a **Packet Sniffer** called **Wireshark**

This article describes how you can check and analyse data passing through a network using a popular packet sniffer called Wireshark.



Packet sniffers or protocol analysers are tools that are commonly used to diagnose network-related problems.

These can also be used by hackers for spying on network user traffic and collecting passwords.

In its simple form, a packet sniffer simply captures all packets of data that pass through a given network interface. Typically, it only captures packets that were intended for the machine in question. However, if placed into promiscuous mode, it is also capable of capturing all packets traversing the network, regardless of destination.

How it works

Packet sniffers work by intercepting and logging network traffic that these can see via the wired or wireless network interface to the computer in use.

On a wired network, what can be captured depends on the structure of the network. A packet sniffer might be able to see traffic on an entire network or only a certain

segment of it, depending on how network switches are configured, placed, etc.

On a wireless network, a packet sniffer can usually only capture one channel at a time, unless the host computer has multiple wireless interfaces that allow for multi-channel capture.

Once raw packet data is captured, the packet sniffing software must analyse it and present it in human-readable form so that those using the software can make sense of it. The people analysing the data can view details of the conversations happening between two or more nodes on the network.

Wireshark

Wireshark is a free and open source packet analyser. It is a computer program that can intercept and log traffic passing over a digital network or part of a network. As data flows across the path, the sniffer captures each packet and, if